FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XII

JEE (Main)-2021

TEST DATE: 20-06-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

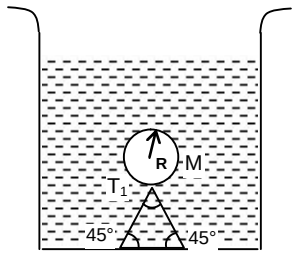
PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. An object is cooled from 75^0C to 65^0C in 2 minutes in a room at 30^0C. The time taken to cool another identical object from 55^0C to 45^0C in the same room in minutes is
   (A) 4
   (B) 5
   (C) 6
   (D) 7

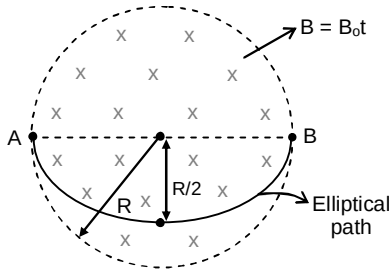
2. A hollow sphere of mass M and radius r is immersed in a tank of water (density ρ_w). The sphere would float if it were set free. The sphere is tied to the bottom of tank by two wires which makes angle 45° with the horizontal as shown in figure. The tension T_1 in the wire is



   (A) $\dfrac{\frac{4}{3}\pi R^3\rho_w g - Mg}{\sqrt{2}}$
   (B) $\frac{2}{3}\pi R^3\rho_w g - Mg$
   (C) $\dfrac{\frac{2}{3}\pi R^3\rho_w g - Mg}{\sqrt{2}}$
   (D) $\frac{4}{3}\pi R^3\rho_w g + Mg$

3. A tunnel is dug along diameter inside the surface of earth and a particle is projected from the centre of tunnel. The minimum velocity of particle such that it escape out from the earth's gravitation field is (Radius of earth = R_e)
   (A) $\sqrt{2gR_e}$
   (B) $\sqrt{\frac{3}{2}gR_e}$
   (C) $\sqrt{3gR_e}$
   (D) $\sqrt{\frac{5}{2}gR_e}$

4. There is a uniform time varying magnetic field in a circular region as shown in the figure. Find out the potential difference across 2 point along an elliptical path as shown in figure.



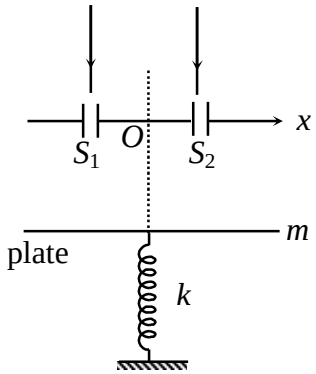
(A) $\frac{\pi R^2}{2}B_o$

(B) $\frac{\pi R^2}{3}B_o$

(C) $\frac{\pi R^2}{4}B_o$

(D) $\frac{\pi R^2}{5}B_o$

5. Two slits S_1 and S_2 lie on the x-axis and symmetric with respect to y-axis are illuminated by a parallel monochromatic light beam of wavelength λ as shown. The distance between slits is d (>> λ). Point O is the mid point of the line S_1S_2 and this point is considered as the origin. The slits are in horizontal plane. The interference pattern is observed on a horizontal plate (acting as screen) of mass m which is connected to one end of a vertical massless spring of spring constant k. The other end of the spring fixed to ground. At t = 0, the plate is at a distance D (>> d) below the plane of slits and spring is in its natural length.



The plate is released from rest from its initial position. The rate by which fringe, width will increase when acceleration of plate is zero, is

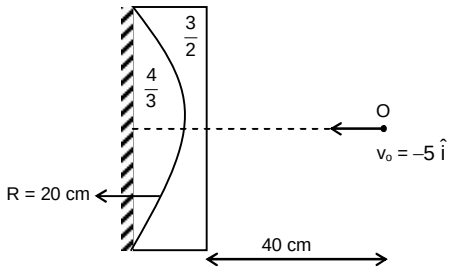
(A) $\frac{\lambda g}{d}\sqrt{\frac{m}{k}}$

(B) $\frac{\lambda g}{3d}\sqrt{\frac{m}{k}}$

(C) $\frac{\lambda g}{4d}\sqrt{\frac{m}{k}}$

(D) $\frac{\lambda g}{2d}\sqrt{\frac{m}{k}}$

6. The lens combination having refractive indices $\frac{4}{3}$ & $\frac{3}{2}$ as shown in the figure, is silvered. The velocity of image will be?



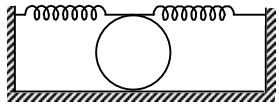
(A) $1.8\,\hat{i}$

(B) $3\,\hat{i}$

(C) $3.6\,\hat{i}$

(D) $5\,\hat{i}$

7. A solid sphere of mass *M* is attached to two massless springs each of spring constant *k.* It can roll without slipping along a horizontal surface. If the system is released after a small stretch in spring, then the time period of oscillations will be :



(A) $2\pi\sqrt{\frac{7M}{2k}}$

(B) $2\pi\sqrt{\frac{7M}{5k}}$

(C) $\pi\sqrt{\frac{7M}{10k}}$

(D) $2\pi\sqrt{\frac{M}{k}}$

8. Water is flowing streamline motion through a horizontal tube. The pressure at a point in the tube is p where the velocity of flow is v. At another point, where the pressure is p/2, the velocity of flow is (density of water $=\rho$) :

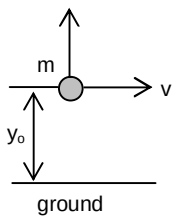
(A) $\sqrt{v^2+\frac{p}{\rho}}$

(B) $\sqrt{v^2-\frac{p}{\rho}}$

(C) $\sqrt{v^2+\frac{2p}{\rho}}$

(D) $\sqrt{v^2-\frac{2p}{\rho}}$

9. A particle of mass m is moving with a velocity of $v=\sqrt{\frac{k}{m}}$ in the horizontal direction at a height y_0 from the ground (y_0 = k/2mg). A variable force F = k/y acts on the particle at t = 0 in the vertical direction where y is the height of the particle from the ground. Find the speed of the particle when its acceleration become zero for the very first time. (k is a constant (N-m)



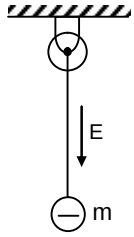
(A) $2\sqrt{\frac{k\,\ell n3}{m}}$

(B) $\sqrt{\frac{2k\,\ell n2}{m}}$

(C) $\sqrt{\frac{2k\,\ell n3}{m}}$

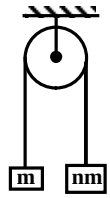
(D) $2\sqrt{\frac{k\,\ell n2}{m}}$

10. A simple pendulum charged negatively to q coulomb oscillates with a time period T in a downward electric field. If the electric field is with-drawn, the new time period is:



(A) = T
(B) > T
(C) < T
(D) none of these

11. Two blocks of masses m and nm are connected by a massless string passing over a frictionless pulley. The value of n for which both the blocks moves with an acceleration of g/10 is



(A) 9/11
(B) 11/9
(C) Both (A) and (B)
(D) None of these

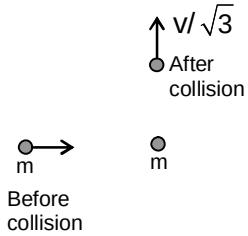
12. The length of one rod is 2.53 cm and that of the other is 1.27 cm. The least count of measuring instrument is 0.01 cm. If two rods are put together end to end, the combined length can be expressed as :

(A) (3.80 ± 0.01) cm
(B) (3.80 ± 0.02) cm
(C) (3.80 ± 0.03) cm
(D) (3.80 ± 0.04) cm

13. If energy (E), velocity (V) and time (T) were chosen as fundamental physical quantities for measurement, then the dimensional formula for mass will be:

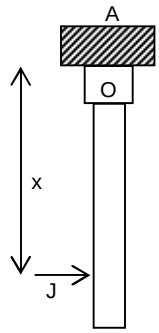
(A) $\left[E^1V^2T^1\right]$
(B) $\left[E^2V^{-2}T^0\right]$
(C) $\left[E^1V^{-2}T^1\right]$
(D) $\left[E^{-1}V^2T^1\right]$

14. A mass m moves with a velocity ν and collides inelastically with another identical mass. After collision, the first mass moves with velocity $\nu/\sqrt{3}$ in a direction perpendicular to the initial direction of motion. Find the speed of the 2nd mass after collision.



(A) $\frac{2\nu}{\sqrt{3}}$
(B) $\frac{\nu}{\sqrt{3}}$
(C) ν
(D) $\sqrt{3}\nu$

15. A uniform rod of length 'l' is pivoted at point 'A'. It is struck by a horizontal force which delivers an impulse 'J' at a distance 'x' from point 'A' as shown in figure, impulse delivered by pivot is zero if 'x' is equal to



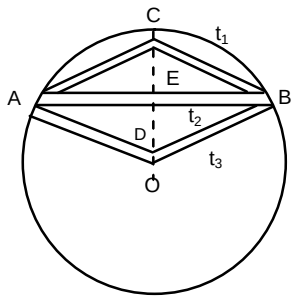
(A) $\frac{l}{2}$

(B) $\frac{l}{3}$

(C) $\frac{2l}{3}$

(D) $\frac{3l}{4}$

16. A particle is dropped in three tunnels ACB, AEB & ADB through earth and takes time to reach at point B respectively t_1, t_2 and t_3.



Then which of the following relation is correct: (D is not the centre of the earth)

(A) $t_1 > t_2 > t_3$

(B) $t_1 < t_2 > t_3$

(C) $t_3 > t_2 > t_1$

(D) $t_2 < t_1 < t_3$

17. A composite string is made up by joining two strings of different masses per unit length, μ and 4μ respectively. The composite string is under the same tension. A transverse wave pulse: y = (6 mm) sin (5t + 40x), where 't' is in seconds and 'x' in meters, is sent along the lighter string towards the joint. The joint is at x = 0. The equation of the wave pulse reflected from the joint is

(A) (2 mm) sin(5t – 40x)

(B) (4 mm) sin(40x – 5t)

(C) –(2 mm) sin(5t – 40x)

(D) (2 mm) sin(5t – 10x)

18. A sphere and a cube of same material and same total surface area are placed in the same evacuated space turn by turn after heating to the same temperature. Find the ratio of their initial rates of cooling in the evacuated space.

(A) $\sqrt{\frac{\pi}{6}} : 1$

(B) $\sqrt{\frac{\pi}{3}} : 1$

(C) $\frac{\pi}{\sqrt{6}} : 1$

(D) $\frac{\pi}{\sqrt{3}} : 1$

19. Two pendulums of same amplitude but time period 3s and 7s start oscillating simultaneously from two opposite extreme positions. After how much time they will be in phase?

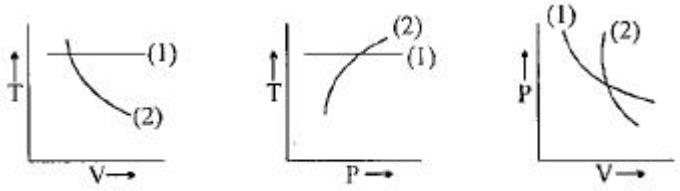
(A) $\frac{21}{8}$s

(B) $\frac{21}{4}$s

(C) $\frac{21}{2}$s

(D) $\frac{21}{10}$s

20.



I II III

If (1) represents isothermal and (2) represents adiabatic, which of the graphs given above in respect of an ideal gas are correct?

(A) (I) and (II)
(B) (II) and (III)
(C) (I), (II) and (III)
(D) Only (III)

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

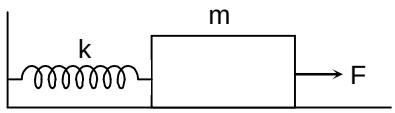
21. A mass of 0.98 kg suspended using a spring of constant K = 300 Nm^{-1} is hit by a bullet of 20 gm moving with a velocity 3.0 ms^{-1} vertically. The bullet gets embedded and oscillates with the mass. Find the amplitude of oscillation.

22. The amplitude of a wave disturbance propagating in the positive x-direction is given by $y = 1/(1 + x^2)$ at time t = 0 and by $y = 1/[1 + (x - 1)^2$ at t = 2 seconds, where x and y are in metres. The shape of the wave disturbance does not change during the propagation. Find the velocity of the wave.

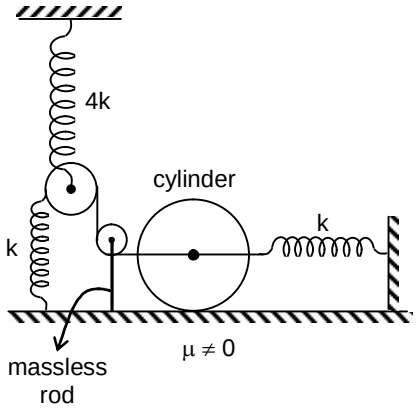
23. A thermometer has an ordinary glass bulb and thin glass tube. The glass bulb is fully filled with 1 mL of mercury and the thin glass tube is completely empty. A temperature change of 1^0C, increases the level of mercury in the thin tube by 3mm. Find the inside diameter of the thin glass tube. $\left(\gamma_{Hg} = 18 \times 10^{-5} /^0 C, \alpha_{glass} = 10^{-5} /^0 C\right)$

24. A projectile is thrown with a velocity of 18 m/s at an angle of 60° with horizontal. The interval between the moments when speed of the projectile is 15 m/s is ($g = 10\ m/s^2$)

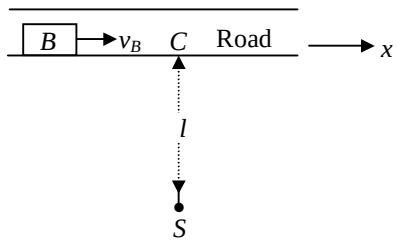
25. A block of mass 'm' is pulled by horizontal constant force
F = 5μmg over a rough surface of coefficient of friction μ as shown. Initially spring was at its natural length, the position where block will finally come to rest is N times of $\frac{5\mu mg}{k}$ where 'N' is



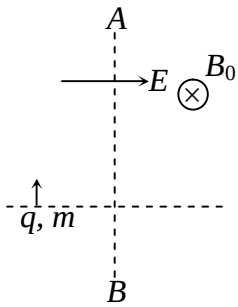
26. A cylinder of mass m = 3 kg and radius R = 2 m is attached to a massless spring-pulley system as shown in the figure and all the springs are in their natural length at t = 0. The friction is sufficient to cause pure rolling of the cylinder. What is the time period (in sec.) of cylinder for small horizontal displacement? (spring constant k = 2 N/m and π = 3.14)



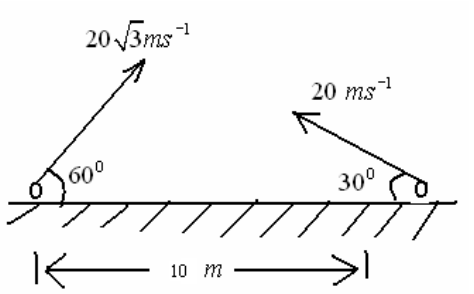
27. A bus B is moving with a velocity v_B in the positive x-direction along a road as shown in the figure. A shooter 'S ' is at a distance l from the road. He has a detector which can detect signals only of frequency 1500Hz. The bus blows horn of frequency 1000 Hz. When the detector detects a signal the shooter immediately shoots towards the road along SC and the bullet hits the bus. Find the velocity of the bullet if velocity of sound in air is v = 340 m/s and $\frac{v_B}{v}=\frac{2}{3\sqrt{3}}$.



28. A uniform electric field *E* is present horizontally along the paper throughout region but uniform magnetic field B_0 is present horizontally (perpendicular to plane of paper in inward direction) right to the line *AB* as shown. A charge particle having charge *q* and mass *m* is projected vertically upward and crosses the line *AB* after time t_0. The speed of projection if particle moves after t_0 with constant velocity. (given *qE* = *mg*) is nt_o. Find the value of 'n'. (g = 9.8 m/s^2)



29. Two particles are projected simultaneously from ground as shown in figure. Find minimum distance in meter is achieved at time (in sec.)



30. Find the binding energy of α-particle in MeV (if m_p = 1.00785 u, m_n = 1.00866 u and m_α = 4.00274 u)

Chemistry

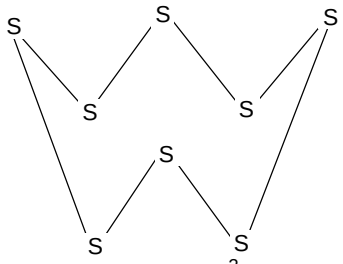
PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which gas shows real behaviour?
(A) 16 g of dioxygen at STP occupies 11.2 L
(B) One gram dihydrogen in 0.5 L flask exerts pressure of 24.63 atm at 300 K
(C) One mole of ammonia at 300 K and 1 atm pressure occupies 22.4 L
(D) 5.6 L of CO_2 at STP is equal to 11 g

32. The oxidation number and hybridization of sulphur in S_8 molecule are respectively:



(A) +2 and sp^2
(B) +2 and sp^3
(C) zero and sp^2
(D) zero and sp^3

33. $X(g) + 2Y(g) \longrightarrow Products$
The rate equation of above reaction is given as follows:
Rate = $k[X][Y]^{0.5}$
What is the unit of rate constant of above reaction?
(A) $mol^{1/2}L^{-1/2}s^{-1}$
(B) $mol^{-1/2}L^{1/2}s^{-1}$
(C) $mol^{3/2}L^{-3/2}s^{-1}$
(D) $mol^{-3/2}L^{3/2}s^{-1}$

34. $HCN(aq) \rightleftharpoons H^+(aq) + CN^-(aq)$
The ionization constant (K_a) of above acid increases by
(A) adding more water
(B) adding more HCN
(C) removing H^+ ions by adding bases
(D) none of these

35. How many effective number of Na^+ ions are present in the unit cell of NaCl if the ions along all the body diagonals of the unit cell are removed?
(A) 4
(B) 3
(C) 2
(D) 1

36. The fraction of chlorine precipitated by $AgNO_3$ solution from $[Cr(NH_3)_5Cl]Cl_2$ is

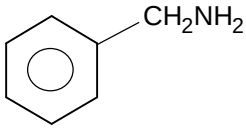
(A) $\frac{1}{2}$

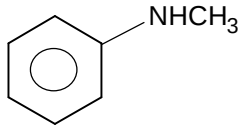
(B) $\frac{1}{3}$

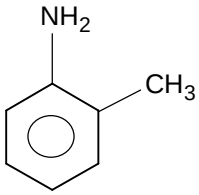
(C) $\frac{2}{3}$

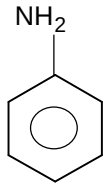
(D) $\frac{1}{4}$

37. Which of the following is the strongest base?

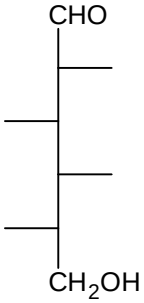
(A) 

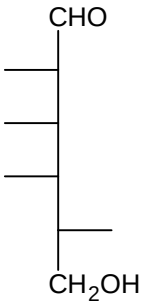
(B) 

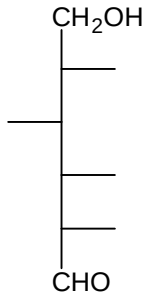
(C) 

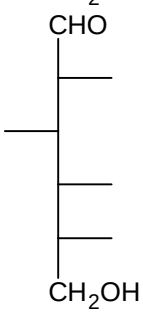
(D) 

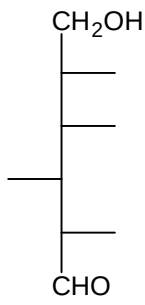
38. Compound which gives same osazone as given by



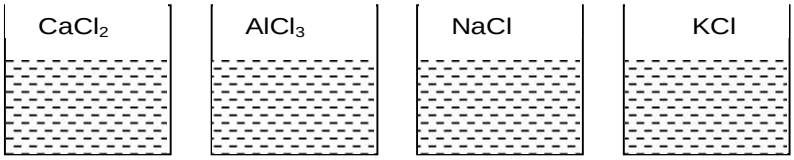
(A) 

(B) 

(C) 

(D) 

39. Equal mass of the following non-volatile substances were added to equal volumes of water in four vessels. Arrange the solutes in decreasing order of their tendency to cause elevation in boiling point of water? [Hint: Assume complete dissociation of salts]



(A) II > I > III > IV
(B) III > II > I > IV
(C) III > IV > I > II
(D) II > III > I > IV

40. Which of the following substance on heating undergoes decarboxylation reaction?

(A) $CH_3CH_2\overset{\overset{O}{\|}}{C}CH_2CH_2COOH$

(B) $CH_3\overset{\overset{O}{\|}}{C}CH_2CH_2CH_2COOH$

(C) $CH_3CH_2CH_2\overset{\overset{O}{\|}}{C}CH_2COOH$

(D) $CH_3CH_2CH_2CH_2\overset{\overset{O}{\|}}{C}COOH$

41. $Ca \xrightarrow{Air} (P + Q) \xrightarrow{H_2O} R + S\uparrow$

$(P + Q) \xrightarrow{CO_2} \underset{\text{White ppt.}}{T}$

$S\uparrow + CO_2 \longrightarrow U + H_2O$

Which of the following is 'U'?

(A) CH_3COONH_4
(B) NH_2CONH_2
(C) NH_2COONH_4
(D) NH_2COOH

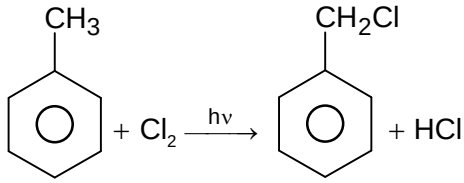
42. Which substance is used for refining of Nickel in Mond's process?

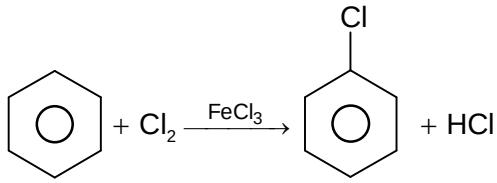
(A) I_2
(B) CO
(C) CO_2
(D) MnO

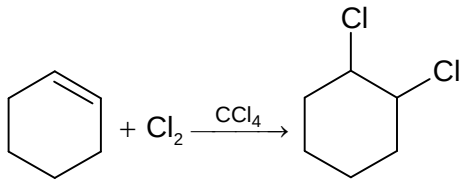
43. Which of the following carbonate needs maximum temperature for decomposition?

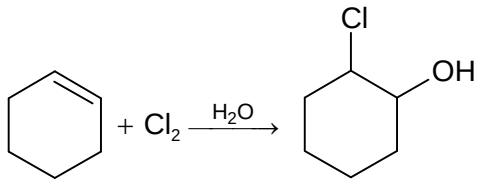
(A) Li_2CO_3
(B) Na_2CO_3
(C) $MgCO_3$
(D) $CaCO_3$

44. Which of the following is an electrophilic substitution reaction?

(A) 

(B) 

(C) 

(D) 

45. The process of 'eutrophication' is due to
(A) increase in concentration of insecticide in water
(B) increase in concentration of fluoride ion in water
(C) the reduction in concentration of the dissolved oxygen in water due to phosphate in water
(D) attack of younger leaves of a plant by peroxyacetyl nitrate

46. The normal boiling point of a liquid 'X' is 500 K. Which of the following statement(s) is/are incorrect for the following process?

$X(\text{liquid}) \longrightarrow X(\text{gas})$

(A) At 500 K and 1 atm pressure, $\Delta G = 0$
(B) At 500 K and 1.5 atm pressure, ΔG = +ve
(C) At 500 K and 0.2 atm pressure, ΔG = -ve
(D) At 520 K and 1 atm pressure, ΔG = +ve

47. The following two equilibria exist simultaneously in a closed container of volume 2 L

$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$

$A_2(g) + C(g) \rightleftharpoons A_2C(g)$

Which of the following change does not affect the concentration of AB(g)
(A) Increase in pressure
(B) Addition of C
(C) Addition of He gas at constant volume
(D) All of these

48. AlF_3 is most soluble in
(A) HF
(B) KF
(C) KHF_2
(D) F_2

49. How much metal is deposited at cathode by passing 965 amp current for 100 sec through molten $AgNO_3$ in presence of unattacked electrodes?
(A) 10.8 g
(B) 108 g
(C) 10.8 mole
(D) 108 mole

50. Which of the following characteristics of lanthanoids decreases uniformly due to lanthanide contraction?
(A) Atomic radius and charge
(B) Ionization enthalpy and ionic radius
(C) Atomic radius and ionic radius
(D) Electronegativity and melting point

SECTION – B
(Numerical Answer Type)

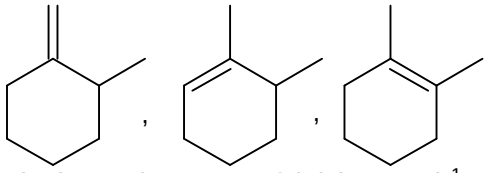
This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. The nuclei of helium and beryllium move with same velocity. If the ratio of their de-Broglie wavelength $\lambda Be^{4+} : \lambda He^{2+}$ is expressed as x : y, what is the value of (y - x)?

52. $X(g) \rightleftharpoons Y(g) + Z(g)$

The equilibrium constant K_P of above reaction is expressed as ℓnK_P = -5. What is the standard free energy change of the reaction in kJ unit at 1000 K?

53. An alkyl chloride(A) gives the following compounds upon treatment with hot ethanolic solution of KOH.



What is the molar mass of (A) in g mol^{-1} unit?

54. What is the molar mass in g mol^{-1} unit of an alkyl chloride which upon treatment with aqueous KOH forms methanol?

55. What is the molar mass of the methyl chloro silane in g mol^{-1} unit which produces cross-linked silicones?
[Atomic mass of Si = 28 g mol^{-1}]

56. What is the kinetic energy of one mole of an ideal gas(helium) at 500 K in joule unit?

57. Yellow $T\ell OH$ turns black on heating at 100°C. If compound(X) is responsible for the black colour. What is its molar mass in g mol^{-1} unit? [At. Mass ot $T\ell$ = 200]

58. A polychloride(X) on NaOH hydrolysis followed by acidification forms HCOOH, what is the molar mass of (X) g mol^{-1} unit?

59. $2A(g) + B(g) \longrightarrow 2C(g)$

Rate of above reaction is 0.4 mol L^{-1} s^{-1}. What is the rate of consumption of A in mol L^{-1} s^{-1} unit?

60. The molar ratio of CH_3COONa and CH_3COOH in a buffer solution is 10 : 1. What is the pH of the buffer?
[K_a of $CH_3COOH = 10^{-4.7}$]

Mathematics PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Locus of midpoint of chords of the circle $x^2 + y^2 = 25$, which are tangents to the circle $x^2 + y^2 - 20x - 20y + 175 = 0$ will be:

(A) $\left(10x + 10y + x^2 + y^2\right)^2 = \left(x^2 + y^2\right)$

(B) $\left(10x + 10y - x^2 + y^2\right)^2 = 100\left(x^2 + y^2\right)$

(C) $\left(10x + 10y + x^2 - y^2\right) = 25\left(x^2 + y^2\right)$

(D) $\left(10x + 10y - x^2 - y^2\right)^2 = 25\left(x^2 + y^2\right)$

62. If combined equation of two sides AB and AC of a triangle is $x^2 - xy - 2y^2 + x + y = 0$ and (1, 1) is orthocenter, then circum – radius of ΔABC is

(A) $\frac{5\sqrt{2}}{3}$

(B) $\frac{10\sqrt{2}}{3}$

(C) $\frac{3\sqrt{2}}{5}$

(D) $\frac{3}{10\sqrt{2}}$

63. Abscissa of two points P and Q on parabola $y^2 = 8x$ are root of equations $x^2 - 17x + 11 = 0$. Tangent at P and Q meet at point T. Distance of T from focus of the parabola is equal to

(A) 5
(B) 6
(C) 7
(D) 8

64. $$f(x) = \begin{cases} \lim\limits_{n\to\infty} \dfrac{x^n \cos x + \cos\left(x + 4x^n\right)}{x^n + 1}, & x \neq 1 \\ k, & x = 1 \end{cases}$$

If $f(x)$ is continuous at x = 1, then

(A) $k = 2\cos 1$

(B) $k = \cos 1$

(C) $k = \cos 1 + \cos 4$

(D) No real value of 'k' exists

65. The area (in sq. units) of the region $A = \left\{(x,y):(x-1)[x] \le y \le 2\sqrt{x},\ 0 \le x \le 2\right\}$, where [t] denotes the greatest integer function, is:

(A) $\frac{8}{3}\sqrt{2}-1$

(B) $\frac{4}{3}\sqrt{2}-\frac{1}{2}$

(C) $\frac{8}{3}\sqrt{2}-\frac{1}{2}$

(D) $\frac{4}{3}\sqrt{2}+1$

66. The integral $\int \frac{\sin^2 x \cos^2 x}{\left(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x\right)^2} dx$ is equal to

(Where C is a constant of integration)

(A) $\frac{-1}{1+\cot^3 x} + C$

(B) $\frac{1}{3\left(1+\tan^3 x\right)} + C$

(C) $\frac{-1}{3\left(1+\tan^3 x\right)} + C$

(D) $\frac{1}{1+\cot^3 + x} + C$

67. If $ab = 1$, $\Delta = \begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$ then minimum value of Δ is

(A) 3
(B) 9
(C) 27
(D) 81

68. If PSP' is focal chord of an ellipse having eccentricity $\frac{3}{4}$. S and S' are its foci and its auxiliary circle is $(x-1)^2 + (y-2)^2 = 25$. If $SP = \frac{140}{93}$, then S'P' is

(A) 6

(B) $\frac{790}{93}$

(C) $\frac{140}{1307}$

(D) 4

69. How many 3×3 matrices M with entries from $\{0,1,2\}$ are there, for which the sum of the diagonal entries of M^TM is 5?

(A) 99
(B) 33
(C) 190
(D) 198

70. If $f(3-x)=f(x)$ and $\int_1^2 f(x)=K$, then $\int_1^2 xf(x)$ is equal to.

(A) K

(B) $\frac{K}{2}$

(C) 2K

(D) $\frac{3}{2}K$

71. If the $(r+1)^{th}$ term in the expansion of $\left(\sqrt[3]{\frac{a}{\sqrt{b}}}+\sqrt{\frac{b}{\sqrt[3]{a}}}\right)^{21}$ has the same power of a and b, then the value of r is

(A) 9
(B) 3
(C) 10
(D) 8

72. A game is played with fair cubic die which has one red side, two blue sides and three green sides. The result is the color of the top side after the dice is rolled. If the die is rolled repeatedly then the probability that second blue result occurs on or before the tenth roll can be expressed in the form of $\frac{3^p-2^q}{3^r}$ where p, q, r are positive integers then the value of $p+r-q$ is

(A) 9
(B) 3
(C) 7
(D) 8

73. The sum of coordinate of point on curve $x^2 = 4y$ which is at minimum distance from the line $y = x - 4$ is equal to

(A) 9
(B) 3
(C) 7
(D) 8

74. Let $a_1, a_2, a_3, \ldots\ldots$ be in A.P. and a_p, a_q, a_r be in G.P. then $a_q : a_p$ is equal to

(A) $\frac{r-p}{q-p}$
(B) $\frac{q-p}{r-q}$
(C) $\frac{r-q}{q-p}$
(D) 1

75. The number of 7 digit numbers whose sum of the digits equals 10 and which is formed by using the digits 1, 2 and 3 only is

(A) 55
(B) 66
(C) 77
(D) 88

76. If $2\sin^2\left(\left(\frac{\pi}{2}\right)\cos^2 x\right) = 1 - \cos(\pi \sin 2x), x \neq (2n+1)\frac{\pi}{2}, n \in I$, then $\cos 2x$ is equal to

(A) $\frac{1}{5}$
(B) $\frac{3}{5}$
(C) $\frac{4}{5}$
(D) 1

77. If $x + iy = \frac{3}{2 + \cos\theta + i\sin\theta}$ then $x^2 + y^2 =$

(A) $4x - 3$
(B) $3x - 4$
(C) $4x + 3$
(D) $3x + 4$

78. The value of $\cos^{-1}\left[\cot\left\{\sin^{-1}\left(\sqrt{\frac{2-\sqrt{3}}{4}}\right)+\cos^{-1}\left(\frac{\sqrt{12}}{4}\right)+\sec^{-1}\left(\sqrt{2}\right)\right\}\right]$ is equal to

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{6}$

79. If A and B are squares matrices such that $A^{2006} = 0$ and AB = A + B, then det(B) =

(A) 0

(B) 1

(C) –1

(D) None of these

80. The minimum area of the triangle whose vertices are A(–1, 1, 2); B(1, 2, 3) and C(t, 1, 1) where t is a real number.

(A) $\frac{\sqrt{3}}{2}$

(B) $\frac{2}{\sqrt{3}}$

(C) $\frac{3\sqrt{3}}{2}$

(D) $\frac{2}{3\sqrt{3}}$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. The curve $y = x^3 + 8x^2 + 40x$ and the line $y = K$ intersect at 3 distinct points. If the x – coordinate of these 3 points form a geometric progression then value of $\frac{K}{10}$ is $(K \in R)$

82. Let 'f' be a differentiable function $\forall x \in R$ where $f(1) = 2$ and $f(3) = 7$ and $f'(x) \geq 2 \forall x \in R$ then minimum integral value of $f(2)$ is

83. The value of $\left(\cot 25^o \cot 35^o + \cos 95^o \cot 145^o + \cot 85^o \cot 155^o\right)$ is equal to

84. Suppose x and y are real numbers and that $x^2 + 9y^2 – 4x + 6y + 4 = 0$ then the minimum value of (4x – 9y) is

85. In a triangle ABC, the equation of the side BC is $2x - y = 3$ and its circumcentre and orthocentre are at (2, 4) and (1, 2) respectively. The value of '$\sqrt{61}\sin B \sin C$' is equal to

86. If the line $y = mx$ bisects the area enclosed by the lines $x = 0, y = 0, x = \frac{3}{2}$ and the curve $y = 1 + 4x - x^2$ for m = $\frac{p}{q}$ (p, q $\in$ N, p & q are co-prime), then the value of 2p–3q =

87. If the tangent to the curve $y = \frac{x}{x^2 - 3}, x \in R, \left(x \neq \pm\sqrt{3}\right)$ at a point $(\alpha, \beta) \neq (0, 0)$ on it is parallel to the line 2x + 6y – 11 = 0 then $|6\alpha + 2\beta|$ =

88. Let z be a complex number such that $\left|\frac{z - i}{z + 2i}\right| = 1$ and $|z| = \frac{5}{2}$. Then the value of $|z + 3i|$ is:

89. If the numbers 330, 486 and 604 divided by a positive number 'p' leaving the remainder 7, 11 and 15 respectively, then the largest value of p is

90. If

$$\lim_{n\to\infty} n^k \int_0^{\frac{1}{n}} x^{x+k-1}\,dx$$

exists and is equal to $f(k), k \in N$, then the value of $\left[\frac{1}{f(5)}\right]$ is equal to (where $[.]$ denotes the greatest integer function)